

CLASSROOM CONTACT PROGRAMME

(ACADEMIC SESSION 2019-2020)

PRE-MEDICAL : ENTHUSIAST COURSE [PHASE-I]

TEST # 06

DATE : 28-11-2019

REVISION TEST

Test Pattern : NEET

INSTRUCTIONS

Do not open this Test Booklet until you are asked to do so

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **180** questions. The maximum marks are **720**.
5. There are **three** parts in the question paper A,B,C consisting of **Physics, Chemistry** and **Biology** having **45 questions** each in Physics & Chemistry and **90 questions** in Biology part of equal weightage. Each question is allotted 4 (four) marks for **correct** response.
6. **One Fourth** mark will be deducted for indicated incorrect response of each question. **No deduction** from the total score will be made if no response is indicated for an item in the Answer Sheet.
7. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** the Answer Sheet. **Use of pencil is strictly prohibited.**
8. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
9. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
10. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
11. **Do not fold or make any stray marks on the Answer Sheet.**

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REVISION TEST # 06

Topics

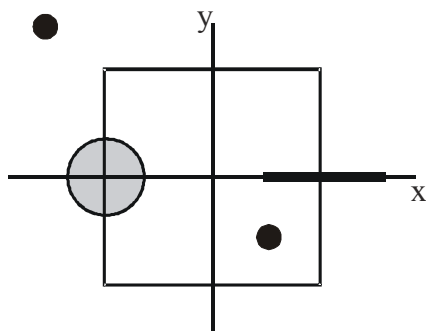
DATE : 28 - 11 - 2019**PRE-MEDICAL ENTHUSIAST COURSE : PHASE - I – PATTERN : [NEET]****PHYSICS** : Electrostatics, Gravitation**CHEMISTRY** : Isomerism**BOTANY** : Plant Diversity (Monera, Protista), Remaining Genetics-I**SPACE FOR ROUGH WORK**

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

1. A disk of radius $a/4$ having a uniformly distributed charge $6C$ is placed in the x - y plane with its centre at $(-a/2, 0, 0)$. A rod of length a carrying a uniformly distributed charge $8C$ is placed on the x -axis from $x = a/4$ to $x = 5a/4$. Two point charges $-7C$ and $3C$ are placed at $(a/4, -a/4, 0)$ and $(-3a/4, 3a/4, 0)$, respectively. Consider a cubical surface formed by six surfaces $x = \pm a/2$, $y = \pm a/2$, $z = \pm a/2$. The electric flux through this cubical surface is :-

- (1) $\frac{-2C}{\epsilon_0}$
(2) $\frac{2C}{\epsilon_0}$
(3) $\frac{10C}{\epsilon_0}$
(4) $\frac{12C}{\epsilon_0}$



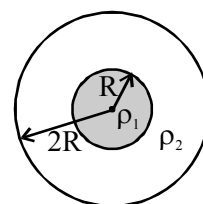
2. If two bodies A and B of equal masses M are situated in air at a distance d and gravitational force between them is F . Now 50% mass is transferred from body A to B and distance between them is reduced to $\frac{d}{2}$. If space around them is now filled with a liquid of specific density 3. The gravitational force now will be :-

- (1) F (2) $3F$
(3) $\frac{3F}{2}$ (4) $\frac{3F}{4}$

3. The density of core of a planet is ρ_1 and that of outer shell is ρ_2 , the radii of the core and that of the planet are R and $2R$ respectively. If acceleration due to gravity at the surface of the planet is same as at

a depth R . The ratio of $\frac{\rho_1}{\rho_2}$ will be :-

- (1) $\frac{3}{8}$ (2) $\frac{8}{3}$
(3) $\frac{3}{7}$ (4) $\frac{7}{3}$



4. For a satellite moving in an orbit around the earth, the ratio of potential energy to kinetic energy is:-

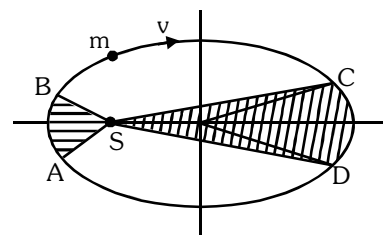
- (1) $\frac{1}{\sqrt{2}}$ (2) 2
(3) $\sqrt{2}$ (4) $\frac{1}{2}$

5. An artificial satellite moving in a circular orbit at a distance h from the centre of the earth has total energy E_0 . Its potential energy is :-

- (1) E_0 (2) $\frac{E_0}{2}$
(3) $2E_0$ (4) $1.5 E_0$

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6. Gravitational acceleration on the surface of a planet is $\frac{\sqrt{6}}{11}g$, where g is the acceleration on the surface of the earth. The average mass density of the planet is $\frac{2}{3}$ times that of earth. If the escape speed on the surface of the earth is 11 km/s, the escape speed on the surface of the planet will be :-
- (1) 3 km/s (2) 6 km/s
(3) 2 km/s (4) 9 km/s
7. The Earth is assumed to be a sphere of radius R . A platform is arranged at a height $2R$ from the surface of the Earth. The escape velocity of a body from this platform is fv , where v is its escape velocity from the surface of the Earth. The value of f is :-
- (1) $\frac{1}{\sqrt{3}}$ (2) $\sqrt{3}$ (3) 3 (4) $\frac{1}{3}$
8. What is the minimum energy required to launch a satellite of mass m from the surface of the earth of mass M and radius R in a circular orbit at an altitude of $2R$.
- (1) $\frac{5GmM}{6R}$ (2) $\frac{2GmM}{3R}$
(3) $\frac{GmM}{2R}$ (4) $\frac{GmM}{3R}$
9. The time period of a satellite in a circular orbit around a planet is independent of
- (1) The mass of the planet
(2) The radius of the planet
(3) The mass of the satellite
(4) All of the above
10. The orbital speed of an artificial satellite close to earth surface is V_0 . The orbital speed of another satellite at a height equal to three times the radius of earth is :-
- (1) V_0 (2) $2V_0$
(3) $0.5V_0$ (4) $4V_0$
11. The figure shows elliptical orbit of a planet m about the sun S . The shaded area SCD is thrice the shaded area SAB . If t_1 is the time for the planet to move from C to D and t_2 is the time to move from A to B then :-
- (1) $t_1 = t_2$
(2) $t_1 > t_2$
(3) $t_1 = 4t_2$
(4) $t_1 = 3t_2$
12. The radii of circular orbits of two satellites A and B of the earth, are $9R$ and R , respectively. If the speed of satellite A is V , then the speed of satellite B will be :-
- (1) V (2) $3V$
(3) $\frac{V}{3}$ (4) $V\sqrt{3}$



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13. A satellite of mass m revolves around the earth (radius R) at a distance x from its surface. If g is the acceleration due to gravity on the surface of earth, then the orbital speed of satellite is :-

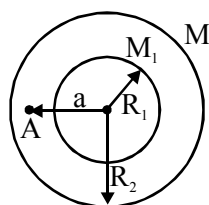
(1) $\sqrt{gx}$ (2) $\sqrt{\frac{gR}{R-x}}$
(3) $\sqrt{\frac{gR^2}{R-x}}$ (4) $\sqrt{\frac{gR^2}{R+x}}$

14. A particle of mass M is situated at the centre of a spherical shell of same mass and radius r . The gravitational potential at a point situated at $\frac{r}{4}$ distance from the centre, will be :-

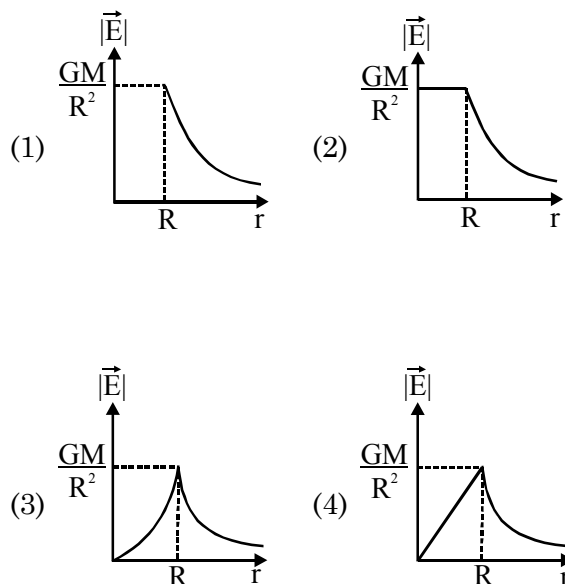
(1) $-\frac{GM}{4}$ (2) $-\frac{3GM}{r}$
(3) $-\frac{4GM}{r}$ (4) $-\frac{5GM}{r}$

15. The figure represents two concentric shells of radii R_1 and R_2 and masses M_1 and M_2 respectively. The gravitational field intensity at point A at a distance "a" from the centre is

(1) $\frac{G(M_1 + M_2)}{a^2}$
(2) $\frac{GM_1}{a^2} + \frac{GM_2}{R_2^2}$
(3) $\frac{GM_1}{a^2}$
(4) Zero



16. Gravitational field due a uniform spherical shell of radius R and mass M varies with r (where r is distance from centre of shell) correctly shown by:-



17. A body of mass m kg starts falling from a point $2R$ above the earth surface. Its kinetic energy when it has fallen to a point upto height R above the earth's surface will be (Mass of earth is M and its radius is R .)

(1) $\frac{GMm}{2R}$ (2) $\frac{GMm}{6R}$
(3) $\frac{2GMm}{3R}$ (4) $\frac{GMm}{3R}$

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18. A small body of superdense material, whose mass is twice the mass of earth but size is very small compared to size of the earth, starts from rest at a height $H \ll R$ above earth's surface and reaches the earth's surface in time t , t is equal to :-
- (1) $\sqrt{\frac{2H}{g}}$ (2) $\sqrt{\frac{H}{g}}$ (3) $\sqrt{\frac{2H}{3g}}$ (4) $\sqrt{\frac{4H}{3g}}$
19. The satellite of mass m revolving in a circular orbit of radius r around the earth has kinetic energy E . Then its angular momentum will be:-
- (1) $\sqrt{\frac{E}{mr^2}}$ (2) $\frac{E}{2mr^2}$
(3) $\sqrt{2Emr^2}$ (4) $\sqrt{2Emr}$
20. Particle-A is projected vertically upward in such a way it reaches maximum height equal to radius of the earth. Another identical particle B projected in same manner reaches up to a maximum height equal to half of the radius of the earth. Find the ratio of power exerted by gravitational force on particle-A and particle-B at the instant they strike to the earth's surface.
- (1) $\sqrt{\frac{3}{2}}$ (2) $\sqrt{\frac{2}{3}}$
(3) $\sqrt{\frac{3}{5}}$ (4) $\sqrt{\frac{5}{3}}$
21. A body of mass m is situated at a distance $4R$ above the earth's surface, where R is the radius of earth. How much minimum energy be given to the body so that it may escape.
- (1) mgR (2) $2mgR$ (3) $\frac{mgR}{5}$ (4) $\frac{mgR}{16}$
22. The height at which acceleration due to gravity will become $\frac{1}{4}$ th of its, value on the surface of earth. (R is radius of earth)
- (1) R (2) $2R$ (3) $3R$ (4) $4R$
23. A satellite-1 is orbiting the earth at a height R above surface of the earth. R is being radius of the earth. If its time period is T . Satellite-2 which is orbiting the earth at a height $7R$ then its time period will be :-
- (1) $8T$ (2) $\frac{T}{8}$ (3) $4T$ (4) $\frac{T}{4}$
24. The mass of planet mars is M and orbital radius of its moon is R . The time period of moon of mars is
- (1) $T^2 = \frac{4\pi^2 R^3}{GM}$
(2) $T^2 = \frac{4\pi^2 GR^3}{M}$
(3) $T^2 = \frac{2\pi R^3 G}{M}$
(4) $T^2 = 4\pi MGR^3$

SPACE FOR ROUGH WORK

25. The moon's radius is $\frac{1}{4}$ that of the earth

and its mass is $\frac{1}{80}$ times that of the earth.

If g represents the acceleration due to gravity on the surface of the earth, its value on the surface of moon will be:-

(1) $\frac{g}{4}$ (2) $\frac{g}{5}$

(3) $\frac{g}{6}$ (4) $\frac{g}{8}$

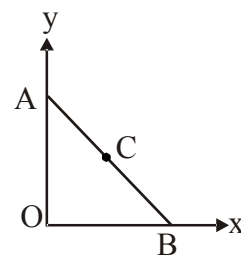
26. An electric dipole has the magnitude of its charge as q and its dipole moment is p . It is placed in a uniform electric field E . If its dipole moment is opposite to the direction of the field, the force on it and its potential energy are respectively :

- (1) $2qE$ and minimum
(2) qE and pE
(3) zero and maximum
(4) qE and maximum

27. A particle of mass 2 mg is having a charge of $2\text{ }\mu\text{C}$. Through what potential difference must it be accelerated, starting from rest, to acquire a speed of 10 m/s ?

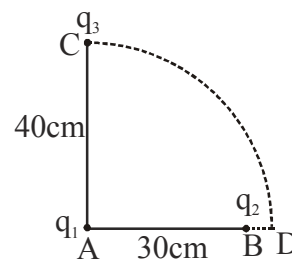
- (1) 5 V (2) 50 V (3) 5 kV (4) 50 kV

28. As per the diagram a point charge $+q$ is placed at the origin O . Work done in taking another point charge $-Q$ from the point A to C is W_1 and from C to B is W_2 , along the straight path AB . coordinates of point A are $(0, a)$ and that of B are $(a, 0)$, point C is mid point between A and B , then



- (1) W_1 and W_2 both are positive
(2) W_1 and W_2 both are negative
(3) $W_1 = -W_2$
(4) W_1 is greater than W_2 in magnitude

29. Two identical charges q_1 and q_2 each at $1\text{ }\mu\text{C}$ are fixed 30 cm apart as shown in the figure. A third identical charge q_3 is moved along the arc of a circle of radius 40 cm from C to D is work done in this process by external agent



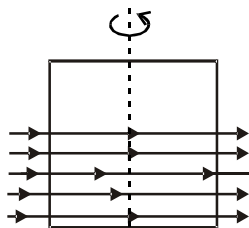
- (1) 144 J (2) 72 J
(3) 288 mJ (4) -144 mJ

30. An electric dipole of moment $\vec{p}$ is released from 90° in uniform electric field $\vec{E}$. The work done by electric field when it is align in the direction of electric field :

- (1) pE (2) $\sqrt{2} pE$
(3) $pE/2$ (4) $2pE$

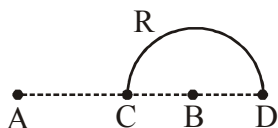
SPACE FOR ROUGH WORK

31. A square surface of side L metres is in the plane of the paper. A uniform electric field $\vec{E}$ (volt/m), also in the plane of the paper, is limited only to the lower half of the square surface (see figure). The change in electric flux if square surface rotates by an angle 90° in SI units associated with the surface is :



- (1) EL^2 (2) $EL^2/2\epsilon_0$
(3) $EL^2/2$ (4) zero

32. Charge $+q$ placed at point A which is a distance $2L$ apart from B and C is the midpoint between A and B. The work done in moving charge $+Q$ along the semicircle CRD is :



- (1) $\frac{qQ}{2\pi\epsilon_0 L}$ (2) $\frac{qQ}{6\pi\epsilon_0 L}$
(3) $-\frac{qQ}{6\pi\epsilon_0 L}$ (4) $\frac{qQ}{4\pi\epsilon_0 L}$

33. Three point charges $+2\mu\text{C}$, $-4\mu\text{C}$ and $+2\mu\text{C}$ are placed at points $(x = 0, y = a, z = 0)$, $(x = 0, y = 0, z = 0)$ and $(x = a, y = 0, z = 0)$ respectively. The magnitude and direction of the electric dipole moment vector of this charge assembly is :

(given $a = 1.41\text{ cm}$)

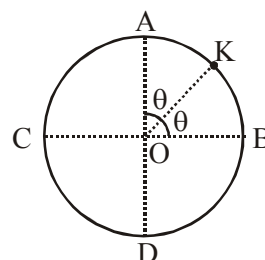
(1) $4 \times 10^{-8}\text{ cm}$ along $\left(\frac{a}{2}\hat{i} + \frac{a}{2}\hat{j}\right)$

(2) $2 \times 10^{-8}\text{ cm}$ along $\left(\frac{a}{2}\hat{i} + \frac{a}{2}\hat{j}\right)$

(3) $4 \times 10^{-8}\text{ cm}$ along $+x$ direction

(4) $2 \times 10^{-8}\text{ cm}$ along $+y$ direction

34. A thin conducting ring of radius R is given a charge $+Q$. The electric field at the centre O of the ring due to the charge on the part AKB of the ring is E . If positive charge of AKB is replaced by same magnitude of negative charge. The net electric field at the centre due to whole ring will be :



- (1) $3E$ along KO
(2) $2E$ along OK
(3) E along KO
(4) $3E$ along OK

SPACE FOR ROUGH WORK

- 35.** An electric field of $(20x\hat{i} + 30y^2\hat{j} + 40z^3\hat{k}) \times 10^{-3} \text{ V/m}$ exists in the space. The electric potential at $A = (3\text{m}, 4\text{m}, 5\text{m})$ is [choose the electric potential at $R = (10\text{m}, 10\text{m}, 10\text{m})$ to be zero, i.e. R as a reference point.]
- (1) 52.01 V (2) 104.02 V
(3) 208.04 V (4) 312.06 V
- 36** The potential field of an electric field $\vec{E} = (y\hat{i} + x\hat{j})$ is
- (1) $V = -xy + \text{constant}$
(2) $V = -(x + y) + \text{constant}$
(3) $V = -(x^2 + y^2) + \text{constant}$
(4) $V = \text{constant}$
- 37.** Two parallel metal plates having charges $+Q$ and $-Q$ face each other at a certain distance d between them. If the plates are now dipped in kerosene oil tank, the electric potential difference between the plates will :
- (1) become zero
(2) increase
(3) decrease
(4) remain same
- 38.** Two parallel large thin metal sheets have equal surface charge densities $(\sigma = 26.4 \times 10^{-12} \text{ C/m}^2)$ of opposite signs. The electric field between these plates is :
- (1) 1.5 N/C
(2) $1.5 \times 10^{-10} \text{ N/C}$
(3) 3 N/C
(4) $3 \times 10^{-10} \text{ N/C}$

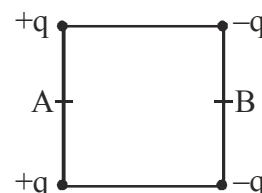
- 39.** Four electric charges $+q, +q, -q$ and $-q$ are placed at the corners of a square of side $2L$ as shown in figure. The electric potential difference between point A , midway between the two charges $+q$ and $+q$, and point B midway between the two charges $-q$ and $-q$ is : $(V_B - V_A)$

(1) $\frac{1}{4\pi\epsilon_0} \frac{4q}{L} \left(\frac{1}{\sqrt{5}} - 1 \right)$

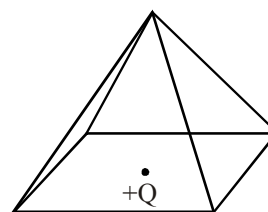
(2) $\frac{1}{4\pi\epsilon_0} \frac{2q}{L} \left(1 + \frac{1}{\sqrt{5}} \right)$

(3) $\frac{1}{4\pi\epsilon_0} \frac{2q}{L} \left(1 - \frac{1}{\sqrt{5}} \right)$

(4) zero



- 40.** A point charge $+Q$ is positioned at the center of the base of a square pyramid as shown. The flux through one of the four identical upper faces of the pyramid is



(1) $\frac{Q}{16\epsilon_0}$

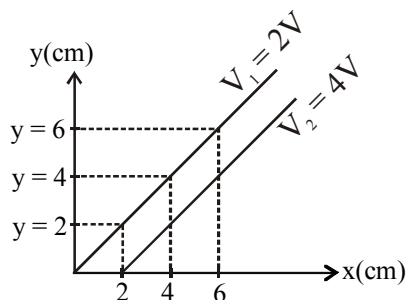
(2) $\frac{Q}{4\epsilon_0}$

(3) $\frac{Q}{8\epsilon_0}$

(4) None of these

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41. Figure shows two equipotential surfaces V_1 and V_2 . The component of electric field in x and y direction will be -



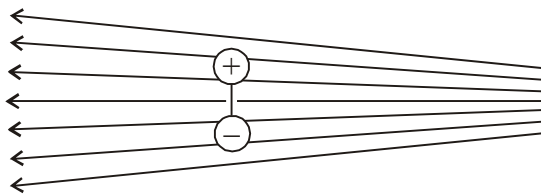
- (1) $E_x \hat{i}$ and $E_y \hat{j}$ (2) $-E_x \hat{i}$ and $E_y \hat{j}$
(3) $E_x \hat{i}$ and $-E_y \hat{j}$ (4) $-E_x \hat{i}$ and $-E_y \hat{j}$
42. A quantity of charge, Q , is distributed uniformly through a sphere of radius R . A smaller sphere, of radius d and concentric with the large sphere, is now removed from it; leaving a spherical cavity with no charge in it. The charge density of the remaining shell has not changed. The electrostatic potential at a distance $r > R$, outside the shell, is

- (1) $\frac{1}{4\pi\epsilon_0} \frac{Q(R^3 - d^3)}{r^3}$ (2) $\frac{1}{4\pi\epsilon_0} \frac{Qr^3}{rd^3}$
(3) $\frac{1}{4\pi\epsilon_0} \frac{QR^3}{rd^3}$ (4) $\frac{1}{4\pi\epsilon_0} \frac{Qd^3}{rR^3}$

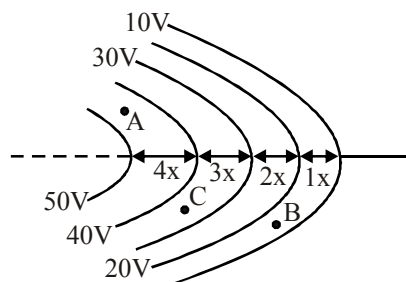
43. A sphere of radius R carries charge density proportional to the square of the distance from the center: $\rho = Ar^2$, where A is a positive constant. At a distance of $R/2$ from the center, the magnitude of the electric field is

- (1) $A/(4\pi\epsilon_0)$ (2) $AR^3/(40\epsilon_0)$
(3) $AR^3/(24\epsilon_0)$ (4) $AR^3/(5\epsilon_0)$

44. When placed in the electric field gradient shown below, the dipole is subjected

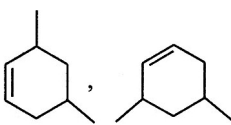


- (1) a clockwise torque and a net linear acceleration
(2) a counterclockwise torque and a net linear acceleration
(3) a clockwise torque and no net linear acceleration
(4) a counterclockwise torque and no net linear acceleration
45. The figure shows lines of constant potential in a region in which electric field is present. The values of the potential are marked in figure. Of the points A, B and C, the magnitude of the electric field is greatest at the point :



- (1) A
(2) B
(3) C
(4) unpredictable

SPACE FOR ROUGH WORK

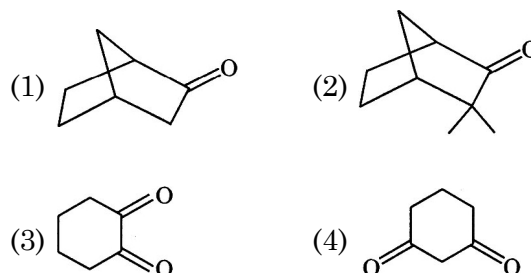
46. An isomer of ethanol is :
 (1) methanol (2) diethyl ether
 (3) acetone (4) dimethyl ether
47. The number of isomers of C_6H_{14} is
 (1) 4 (2) 5 (3) 6 (4) 7
48. 
- Relation between the above compounds is :
 (1) Position isomers
 (2) Chain isomers
 (3) Identical
 (4) Functional isomer
49. Total number of position isomers of dimethyl cyclohexane.
 (1) 2 (2) 3 (3) 4 (4) 5
50. Which type of isomerism is observed between I and II?

$$CH_3-CH_2-\overset{\overset{O}{\parallel}}{C}-OCH_3, CH_3-\overset{\overset{O}{\parallel}}{C}-OC_2H_5$$

(I)
(II)

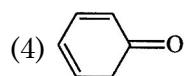
 (1) Functional isomerism
 (2) Metamerism
 (3) Position isomerism
 (4) Stereoisomerism
51. How many different structural isomers exist for C_3H_6O in which no atom is sp^2 or sp -hybridised?
 (1) 2 (2) 5 (3) 4 (4) 3
52. How many structural isomers of tertiary amines corresponding to molecular formula $C_6H_{15}N$ are possible?
 (1) 4 (2) 5 (3) 6 (4) 7

53. Among the following compounds, one that will not show keto-enol tautomerism is

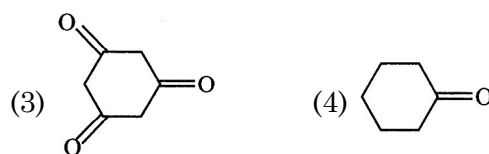


54. Maximum enolisation takes place in

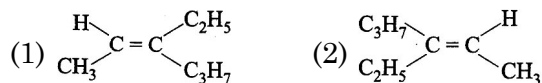
- (1) CH_3COCH_3
 (2) CH_3COCH_2CHO
 (3) $CH_3COCH_2COCH_3$



55. Tautomerism is not exhibited by



56. The Z-configuration in the following is



57. An alkene exists as geometrical isomers. The minimum number of carbon atoms in the molecules is

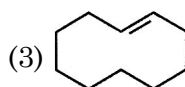
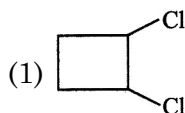
- (1) 1 (2) 2 (3) 3 (4) 4

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58. The number of isomers for the compound with molecular formula $C_2BrClFI$ is
(1) 3 (2) 4 (3) 5 (4) 6

59. Geometrical isomerism is possible in
(1) Acetone-oxime
(2) Isobutene
(3) Acetophenone-oxime
(4) Benzophenone-oxime

60. Which will form geometrical isomers?



(4) All of these

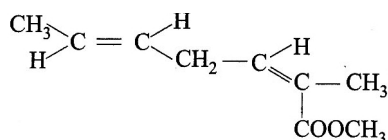
61. Which type of isomerism cannot be shown by benzaldoxime?

- (1) Optical isomerism
(2) Functional group isomerism
(3) Geometrical isomerism
(4) Configuration isomerism

62. How many cyclic isomers (structural and geometrical only) exist for C_5H_{10} ?

- (1) 3 (2) 5 (3) 6 (4) 8

63. The correct stereochemical name of

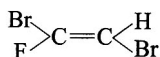
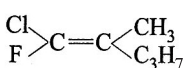
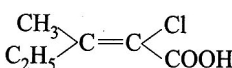
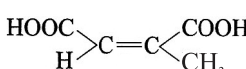


- (1) Methyl 2-methylhepta (2E, 5E) dienoate
(2) Methyl-2-methylhepta (2Z, 5Z) dienoate
(3) Methyl-2-methylhepta (2E, 5Z) dienoate
(4) Methyl-2-methylhepta (2Z, 5E) dienoate

64. Which compound can show geometrical isomerism?

- (1) $CH_3CH = C(CH_3)_2$
(2) $CH_3CH = CH_2$
(3) $CH_3CH = CHCH_3$
(4) $(CH_3)_2C = C(CH_3)_2$

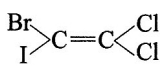
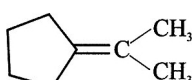
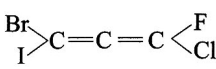
65. The 'Z'-isomer is :

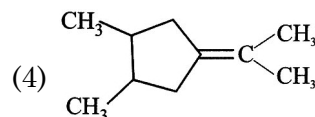
- (1) 
(2) 
(3) 
(4) 

66. The number of geometrical isomers for $CH_3 - CH = CH - CH = CH - CH = CH_2$ is

- (1) 2 (2) 4 (3) 6 (4) 8

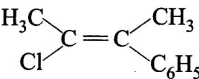
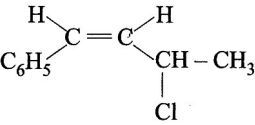
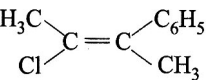
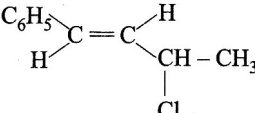
67. Which of the following compound can show geometrical isomerism?

- (1) 
(2) 
(3) 



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68. The correct stereochemical formula of trans-3-chloro-1-phenylbut-1-ene is

- (1) 
- (2) 
- (3) 
- (4) 

69. How many optically active stereoisomers are possible for butane-2,3-diol?

- (1) 1 (2) 2 (3) 3 (4) 4

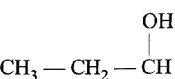
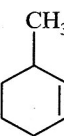
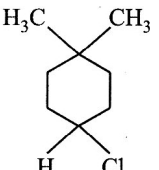
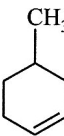
70. The smallest alcohol which exhibit optical activity is

- (1) tert-Butyl alcohol (2) Butan-1-ol
(3) Pentan-2-ol (4) Butan-2-ol

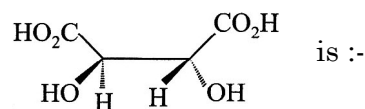
71. What is the molecular formula for the alkane of smallest molecular weight which possesses a stereogenic centre?

- (1) C₇H₁₆ (2) C₆H₁₄ (3) C₅H₁₂ (4) C₈H₁₈

72. Which of the following compound is optically inactive?

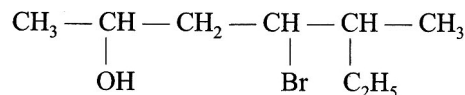
- (1)  (2) 
- (3)  (4) 

73. The absolute configuration of



- (1) S, S (2) R, R
(3) R, S (4) S, R

74. Number of chiral carbon present in the following compound:

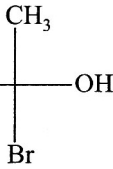
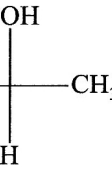
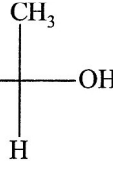
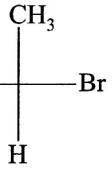


- (1) 2 (2) 3 (3) 4 (4) 5

75. The number of stereoisomers obtained by bromination of trans-2-butene is

- (1) 1 (2) 2 (3) 3 (4) 4

76. Which of the following compound has 'S' configuration?

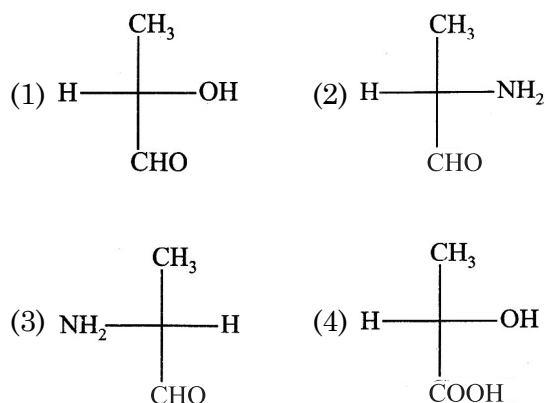
- (1)  (2) 
- (3)  (4) 

77. Enantiomers are

- (1) molecules that have at least one chiral carbon
(2) non-superimposable molecules that are mirror images of one another
(3) non-superimposable constitutional isomers
(4) non-superimposable molecules

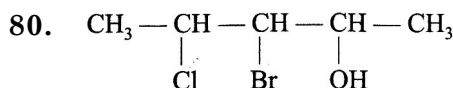
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78. Which of the following compound has 'D' configuration?



79. Which of the following compounds is not chiral?

- (1) 3-chloro-2-methyl pentane
- (2) 2-chloropentane
- (3) 1-chloro-2-methyl pentane
- (4) 1-chloropentane



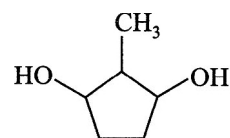
Total number of stereoisomers in above compound is :

- (1) 6 (2) 4 (3) 8 (4) 16

81. A hydrocarbon X is optically active. X upon hydrogenation gives an optically inactive alkane Y. Which of the following pair of compounds can be X and Y respectively?

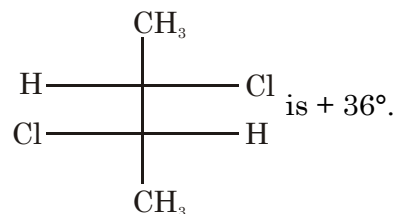
- (1) 4-methyl-2-hexene and 3-methyl hexane
- (2) 2-methyl-1-butene and 2-methyl butane
- (3) 4-methyl cyclopentene and methyl cyclopentane
- (4) 3-methyl-1-pentene and 3-methyl pentane

82. How many different stereoisomers exist for the compound below?

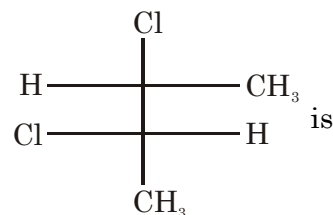


- (1) 2 (2) 8
(3) 6 (4) 4

83. If optical rotation produced by



then that produced by



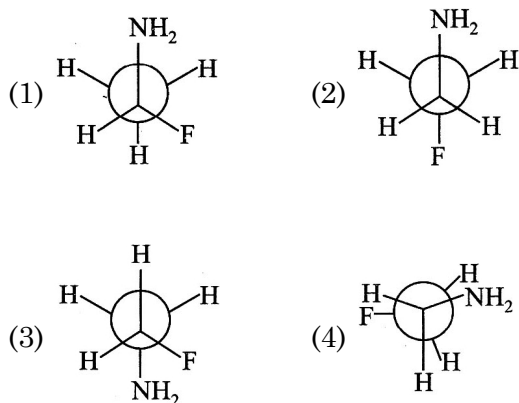
- (1) -36° (2) 0°
(3) $+36^\circ$ (4) $+18^\circ$

84. The number of stereoisomers possible for a compound of the molecular formula $\text{CH}_3 - \text{CH} = \text{CH} - \text{CH}(\text{OH}) - \text{Me}$ is

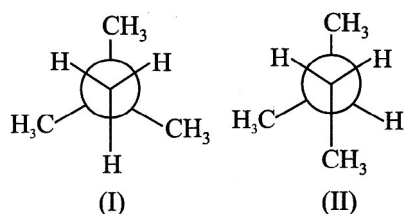
- (1) 3 (2) 2
(3) 4 (4) 6

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85. The most stable conformation of 2-fluoroethan-1-amine is:



86. The two structures I and II represent:



- (1) Conformational isomers
- (2) Stereoisomers
- (3) Constitutional isomers
- (4) Identical

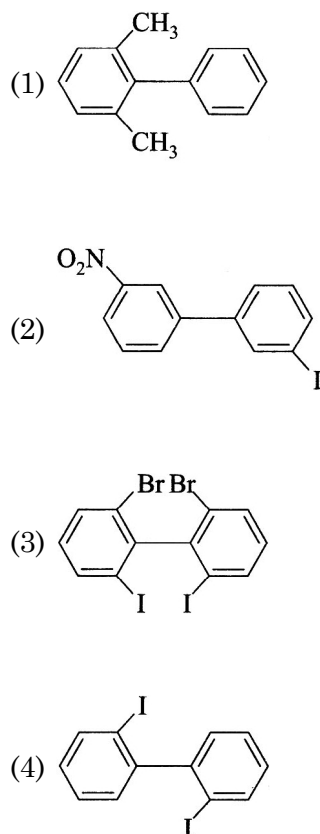
87. The correct energy gradation of different conformers is

- (1) Staggered > skew > eclipsed
- (2) Skew > staggered > eclipsed
- (3) Eclipsed > skew > staggered
- (4) Skew > staggered > eclipsed

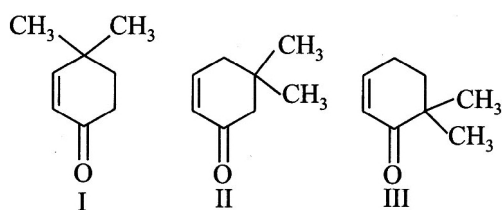
88. Most stable form of cyclohexane is

- (1) Boat
- (2) Chair
- (3) Skew
- (4) eclipsed

89. Which of the following biphenyls is optically active?



90. Given :

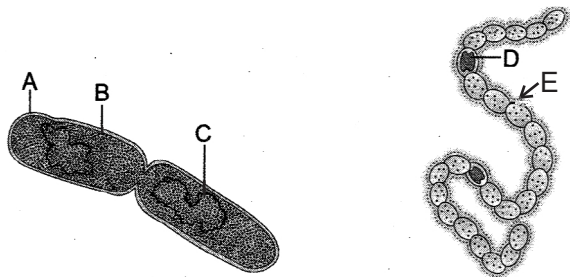


Which of the given compounds can exhibit tautomerism?

- (1) II and III
- (2) I, II and III
- (3) I and II
- (4) I and III

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91. Identify the labellings in the following figures.



- (1) A-Cell wall, B-Cell membrane, C-Heterocyst, D-NA, E-Mucilaginous sheath
- (2) A-Cell wall, B-Cell membrane, C-DNA, D-Heterocyst, E-Mucilaginous sheath
- (3) A-Mucilaginous sheath, B-Cell membrane, C-DNA, D-Heterocyst, E-Cell wall
- (4) A-Cell membrane, B-Cell wall, C-DNA, D-Heterocyst, E-Mucilaginous sheath

92. Which of the following kingdoms has no well defined boundaries?

- (1) Monera
- (2) Protista
- (3) Fungi
- (4) Metaphyta and Metazoa

93. In which of the following, the cell wall has stiff cellulose plate on the outer surface?

- (1) Dinoflagellates
- (2) Desmids
- (3) Diatoms
- (4) Euglenoids

94. Select the correct combination of the statements regarding the characters of certain organisms :

- (a) Methanogens are archaeobacteria which produce methane in marshy areas.
 - (b) Chemosynthetic autotrophic bacteria synthesise cellulose from glucose.
 - (c) Mycoplasma have cell wall and survive in presence of oxygen.
 - (d) Nostoc perform N_2 fixation in heterocyst cell of filament
- (1) a and b
 - (2) a and d
 - (3) b and c
 - (4) c and d

95. Which statements are right for euglena :

- (a) They have unequal flagella.
- (b) They represent only autotrophic mode of nutrition.
- (c) They are found in fresh water.
- (d) They have pigments similar to angiosperms.
- (e) They may obtain food by predating on other small organisms.

- (1) b, c & e
- (2) a, b & d
- (3) a, b, c & e
- (4) a, c, d & e

96. Match the following Column I and Column II :

Column I

Column II

- (a) Slime mould
 - (b) Dinoflagellates
 - (c) Chrysophytes
- (i) Diatoms
 - (ii) Yellow and blue colour
 - (iii) Gonyaulax
 - (iv) Saprophytic
 - (v) Represent soap box like structure
 - (vi) Fruiting body formation

- (1) a - ii, iii b - iv, vi c - i, v
- (2) a - iv, vi b - ii, iii c - i, v
- (3) a - iii, vi b - i, iv c - ii, v
- (4) a - v, vi b - i, iv c - ii, iii

97. How many given characters of organisms are used as main criteria for five kingdom classification :

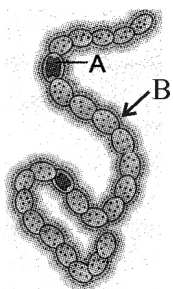
- (i) Cell structure
 - (ii) Thallus organisation
 - (iii) Reproduction
 - (iv) Mode of nutrition
 - (v) Venation of leaf in plants
 - (vi) Phylogenetic relationship
 - (vii) Pigments of leaf perform photosynthesis.
- (1) 4
 - (2) 5
 - (3) 2
 - (4) 3

98. A pair of bacteria involved in two steps during oxidation of ammonia into nitrate :

- (1) Clostridium and Azotobacter
- (2) Nitrosomonas and nitrobacter
- (3) Rhizobium and Pseudomonas
- (4) Xanthomonas and Rhodospirillum

- 99.** Which of the following is incorrect statement :-
- (1) Entire circular DNA of donor cell enter into the recipient cell during transduction of bacteria
 - (2) Transduction is method of genetic recombination in some bacteria
 - (3) Conjugation tube in bacterial conjugation in genetic recombination prepared by Pili
 - (4) Extra chromosomal DNA in bacteria called plasmid
- 100.** Select wrong statement for Heterocyst :-
- (1) They are found in *Nostoc* and *Anabaena*
 - (2) They are found in chlorella
 - (3) They have nitrogenase enzyme to perform N_2 fixation
 - (4) Mostly cyanobacteria found in water of paddy field and perform N_2 fixation in heterocyst
- 101.** Most common method of reproduction in bacteria during favourable condition is :-
- (1) Spore formation
 - (2) Conjugation
 - (3) Budding
 - (4) Binary fission
- 102.** The main difference between gram positive and gram negative bacteria in the composition of
- (1) Cilia
 - (2) Cell wall
 - (3) Cell membrane
 - (4) Cytoplasm
- 103.** Moneran lack of cell wall are
- (1) Eubacteria
 - (2) Archaeobacteria
 - (3) Mycoplasma
 - (4) Cyanobacteria
- 104.** Mycoplasma are smallest living cell known and can survive without
- (1) Cell membrane
 - (2) Oxygen
 - (3) Heterotrophic nutrition
 - (4) Genetic material
- 105.** Which type of genetic material found in bacteria?
- (1) Double stranded circular DNA without histone protein
 - (2) Double stranded linear DNA without histone protein
 - (3) Single stranded DNA with high GC content
 - (4) Low molecular weight single stranded RNA
- 106.** *Rhodospirillum* is a
- (1) Purple sulphur bacteria
 - (2) Purple nonsulphur bacteria
 - (3) Green sulphur bacteria
 - (4) Green nonsulphur bacteria
- 107.** Purple sulphur bacteria
- (1) Are photo organotroph
 - (2) Possess chlorophyll a and phycobillins
 - (3) Use H_2S as H^+ donor
 - (4) Perform oxygenic photosynthesis
- 108.** *E. Coli* and *Anabaena* resemble in the presence of
- (1) Heterocyst
 - (2) Oxygenic photosynthesis
 - (3) Peptidoglycan cell wall
 - (4) Chromatophore
- 109.** The process of genetic recombination in which DNA is transferred from a bacterium to another bacterium by physical contact is known as
- (1) Transformation
 - (2) Conjugation
 - (3) Specialised transduction
 - (4) Generalised transduction

110. Identify the given diagram with correct label.



- (1) Nostoc A-Heterocyst B-Cell wall
- (2) Nostoc A-Heterocyst B-Mucilaginous sheath
- (3) Nostoc A-Vegetative cell B-Cell wall
- (4) Actinomycetes A-Endospore B-Glycocalyx

111. If a bacterium grow only in anaerobic environment and die in the presence of oxygen then it is called as

- (1) Obligate aerobe
- (2) Facultative aerobe
- (3) Obligate anaerobe
- (4) Facultative anaerobe

112. Most of the bacteria are :-

- (1) Autotrophs
- (2) Myxotrophs
- (3) Heterotrophs
- (4) None of these

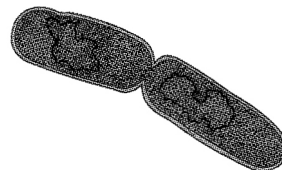
113. During transduction in bacteria, a small segment of double stranded DNA is transferred from donor to recipient bacterium by

- (1) Conjugation tube
- (2) Bacteriophage
- (3) Fungi
- (4) Prions

114. During conjugation between F^+ and F^- bacterial cells the F^- cell is converted to

- (1) F^+
- (2) Hfr
- (3) F^-
- (4) Meta male

115. Select the incorrect option w.r.t. binary fission method in bacteria shown below in the diagram.



- (1) Occurs under favourable condition
- (2) Involves DNA replication
- (3) Sort of sexual reproduction
- (4) The daughter cells formed are genetically identical

116. Characteristic photosynthetic pigments present in BGA are

- (1) Chlorophyll a, Chlorophyll b
- (2) Chlorophyll a, phycocyanin and phycoerythrin
- (3) Chlorophyll c, Phycobilins
- (4) Chlorophyll c, phycocyanin and phycoerythrin

117. Which of the following is not related to BGA?

- (1) Nitrogen Fixation
- (2) Cyanobacteria
- (3) Oxygenic photosynthesis
- (4) Nonoxygenic photosynthesis

118. Which of the following played a significant role in the evolution of O_2 on earth?

- (1) Archaeobacteria
- (2) BGA
- (3) Green Algae
- (4) Protistan algae

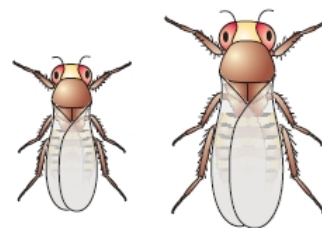
119. Edible fast growing cyanobacteria used as source of single cell protein (SCP) is

- (1) *Spirulina*
- (2) *Nostoc*
- (3) *Anabaena*
- (4) *Microcystis*

- 120.** Algal bloom in the polluted water bodies are often formed by
- (1) Blue green algae
 - (2) Sulphur bacteria
 - (3) Nitrifying bacteria
 - (4) Archaeobacteria
- 121.** Which of the following organisms are responsible for the production of biogas from the dung of cows and buffaloes?
- (1) Methanogens
 - (2) Halophiles
 - (3) Thermoacidophiles
 - (4) Actinomycetes
- 122.** Organisms with soap box like structure are
- (1) Diatoms
 - (2) Dinoflagellates
 - (3) Consumer decomposer protist
 - (4) Euglenoids
- 123.** A longitudinal and a transverse flagellum are feature of organisms
- (1) Diatoms
 - (2) Sporozoans
 - (3) Slime moulds
 - (4) Dinoflagellates
- 124.** Which organisms undergo such a rapid multiplication that they make the sea appear red (called red tides)?
- (1) *Navicula*
 - (2) *Trichodesmium*
 - (3) *Gonyaulax*
 - (4) *Chlamydomonas*
- 125.** Cell wall is indestructible in diatoms due to presence of
- (1) Pectin
 - (2) Cellulose
 - (3) Silica
 - (4) Chitin
- 126.** Diatomaceous earth is used for all except
- (1) Polishing
 - (2) Filtration of oil and syrup
 - (3) Conductor in refrigerator
 - (4) Soundproofing
- 127.** are the chief producers in the oceans.
- (1) Diatoms
 - (2) Dinoflagellates
 - (3) BGA
 - (4) Euglenoids
- 128.** Bacteria reproduce by binary fission during A and endospore formation during B condition
- | A | B |
|------------------|--------------|
| (1) Favourable | Favourable |
| (2) Unfavourable | Favourable |
| (3) Favourable | Unfavourable |
| (4) Unfavourable | Unfavourable |
- 129.** Endospores are
- (1) Thin walled and Metabolically inactive
 - (2) Thick walled and stress intolerant
 - (3) Thick walled and resistant
 - (4) Metabolically active reproductive structure of bacteria
- 130.** Select the incorrect match w.r.t. reproduction in bacteria.
- (1) Transformation - Griffith
 - (2) Conjugation - Done without F-Plasmid
 - (3) Endospore formation - Reproduction without multiplication
 - (4) Binary fission - Most common method of multiplication
- 131.** Cyanobacteria do not possess/ show
- (1) Flagella
 - (2) Gene recombination
 - (3) 70S Ribosome
 - (4) Lamellosome
- 132.** Heterocyst present in cyanobacteria is specialised for N_2 fixation and it possess
- (1) Thick walled
 - (2) Oxygen
 - (3) Nitrogenase
 - (4) Both (1) and (3)
- 133.** Which is true for archaeobacteria?
- (1) All are Methanogens
 - (2) All are chemoautotroph
 - (3) Oldest living fossils
 - (4) All are fossils

- 134.** In five kingdom classification, multicellular Eukaryotic organism with autotrophic mode of nutrition are included in
- (A) Animalia (B) Fungi
(C) Plantae (D) Protista
- (1) Only (C) is correct
(2) Both (A) and (B) are correct
(3) Only (B) is correct
(4) All are correct
- 135.** Free living nitrogen fixing bacteria are found in
- (1) Air
(2) Soil
(3) Root nodules
(4) All of these
- 136.** The function of mesosome in prokaryotes
- (1) Respiration
(2) DNA replication
(3) Both (1) and (2)
(4) Nitrogen fixation
- 137.** Which of the following can utilize light energy for the synthesis of ATP
- (1) Slime mould
(2) Cephaleuros
(3) Halophiles
(4) Thermoacidophiles
- 138.** Harmful activity of BGA is :-
- (1) N₂-Fixation
(2) Increase fertility of soil
(3) Water bloom
(4) Increase alkalinity of soil
- 139.** In bacteria respiratory enzymes are present in
- (1) Cell wall
(2) Mitochondria
(3) Plasma membrane
(4) Nucleoid

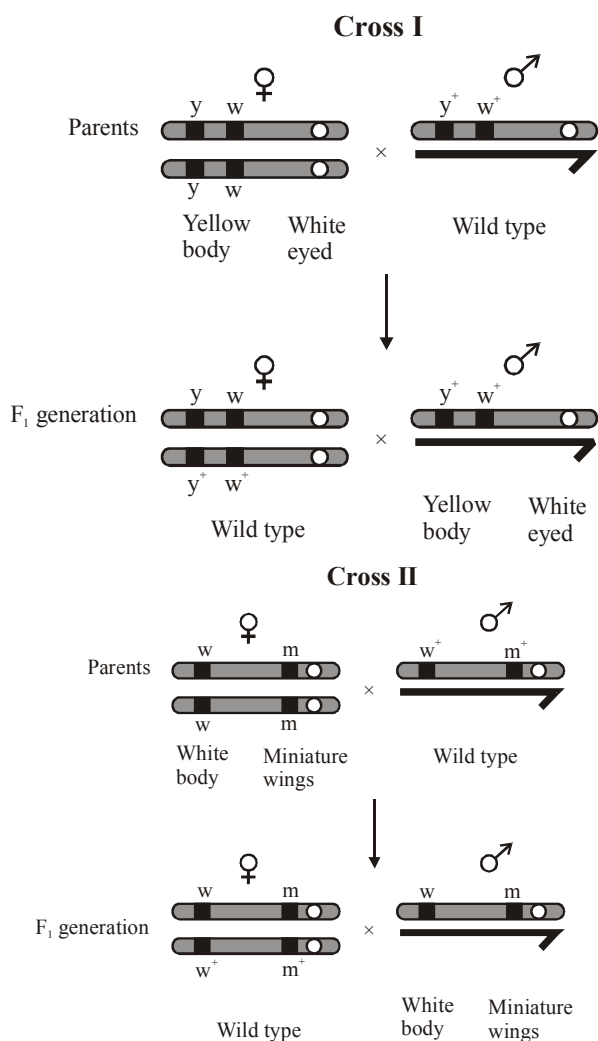
- 140.** On the basis of their shape bacteria are grouped under _____ categories.
- (1) Two (2) Three
(3) Six (4) Four
- 141.** Example of methanogen archaebacteria
- (1) Methanococcus
(2) BGA
(3) Slime mould
(4) Mycoplasma
- 142.** Archaebacteria found in hot springs are known as
- (1) Methanogens
(2) Thermoacidophiles
(3) Cryophiles
(4) Halophiles
- 143.** Mycoplasma is not affected by penicillin due to
- (1) Pleomorphic nature
(2) Presence of linear DNA
(3) Lack of cell wall
(4) Presence of peptidoglycan cell wall
- 144.** Given below is, the figure of *Drosophila melanogaster*.



It was found to be very suitable for experimental verification of chromosomal theory of inheritance by Morgan, because.

- (1) A single mating produces two young flies
(2) Smaller female is easily recognisable from larger male
(3) It completes life cycle in about two weeks
(4) It reproduces parthenogenetically

145. The experiment shown in the figure has been carried out by Morgan to show the phenomenon of linkage & recombination. If in Cross I, genes are tightly linked and in Cross II, genes are loosely linked then what will be percentage of recombinants produced in Cross I and Cross II respectively in F_2 gene ration ?



- (1) 98.7% and 62.8% (2) 1.3% and 37.2%
(3) 37.2 and 1.3% (4) 62.8% and 98.7%

146. Given below is the karyotype of *Drosophila*.

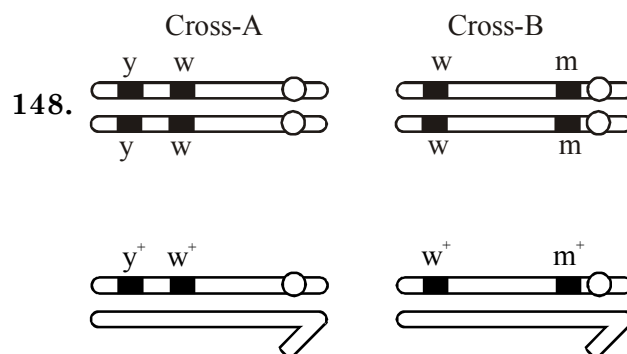


The sex of such *Drosophila* should be –

- (1) Superfemale (2) Intersex
(3) Sterile male (4) Supermale

147. Crossing over is advantageous because it brings about

- (1) variation (2) linkage
(3) inbreeding (4) stability



Identify cross A and B in Morgan experiment:-

	Cross-A	Cross-B
(1)	Tightly linked genes	Loosely linked genes
(2)	Linkage	Independent assortment
(3)	Independent assortment	Linkage
(4)	More recombination	Less recombination

149. In a linear chromosome, map distance between four loci are $ab = 12$, $bc = 4$, $ad = 2$, $ac = 8$. The distance between c and d will be -

- (1) 10% (2) 6%
(3) 10% or 6% (4) 3% or 6%

150. Two genes a and b are linked and show 30% recombination. If ++ / ++ individual married with ab / ab then the types and proportion of gametes in F_1 will be -

- (1) ++ 35% : ab : 35% : + a 15% : +b 15%
(2) ++ 15% : ab : 15% : + a 35% : +b 35%
(3) ++ 35% : ab : 15% : + a 35% : +b 15%
(4) ++ 15% : ab : 35% : + a 15% : +b 35%

151. Coupling and Repulsion theory produced by:-

- (1) Morgan (2) Bateson
(3) Muller (4) De vries

- 152.** Morgan and his group found that when genes were grouped on the same chromosome, some genes were very tightly linked and showed—
 (1) Very low recombination
 (2) Higher recombination
 (3) No recombination
 (4) 100% parental combination
- 153.** Which statement is not true for *Drosophila melanogaster*—
 (1) They complete their life cycle about two weeks
 (2) Single mating produce large number of progeny flies
 (3) It has few hereditary variation that can be seen with high power microscope
 (4) It has clear differentiation of the sex
- 154.** Which of the following compounds are decomposed during putrefaction
 (1) Proteins (2) Fats
 (3) Carbohydrates (4) None
- 155.** Nucleoid is present in
 (1) Bacteria (2) Virus
 (3) Slime moulds (4) Fungi
- 156.** Which of the following can be called as genophore
 (1) A gene of *Drosophila*
 (2) Genes of *Neurospora*
 (3) Bacterial DNA
 (4) Eukaryotic DNA
- 157.** Purple sulphur bacteria are
 (1) Photolithoautotroph
 (2) Photo-organotroph
 (3) Chemolithotroph
 (4) Chemo-organotrophs
- 158.** When cocci are present in the bunch, they are called
 (1) *Diplobacilli* (2) *Streptococci*
 (3) *Staphylococci* (4) *Streptobacilli*
- 159.** An episome in bacteria is
 (1) Milk factor
 (2) Co-factor
 (3) Sex factor or Fertility factor
 (4) None of the above
- 160.** Branched chain lipids occur in the cell membranes of
 (1) *Archaeobacteria*
 (2) *Mycoplasma*
 (3) *Actinomycetes*
 (4) *Streptomyces*
- 161.** Bacteria are found to be primitive organism because they
 (1) Are small, microscopic which are not seen with naked eye
 (2) Cause serious diseases to human being, domesticated animals and crop plants
 (3) Produce endospores which are very resistant to adverse conditions
 (4) Possess incipient nucleus and show amitotic division
- 162.** Pili in bacteria represent
 (1) Extra-chromosomal genetic element
 (2) Protoplasmic outgrowths of donor cells
 (3) Large flagella
 (4) Special bacterial cilia
- 163.** A free living anaerobic bacterium capable of N_2 fixation in soil is
 (1) *Rhizobium* (2) *Azotobacter*
 (3) *Streptococcus* (4) *Clostridium*
- 164.** Which of the following is free-living aerobic non- photosynthetic nitrogen fixing bacterium
 (1) *Rhizobium* (2) *Azotobacter*
 (3) *Nostoc* (4) *Streptomyces*
- 165.** Nitrifying bacteria, *Nitrosomonas* and *Nitrobacter*
 (1) Convert (oxidise) ammonia or ammonium compounds into nitrates
 (2) Convert nitrate into nitrogen
 (3) Convert nitrogen into nitrates
 (4) Convert carbon dioxide into carbohydrates
- 166.** Little leaf of brinjal is caused by
 (1) Virus (2) *Mycoplasma*
 (3) Fungus (4) Algae

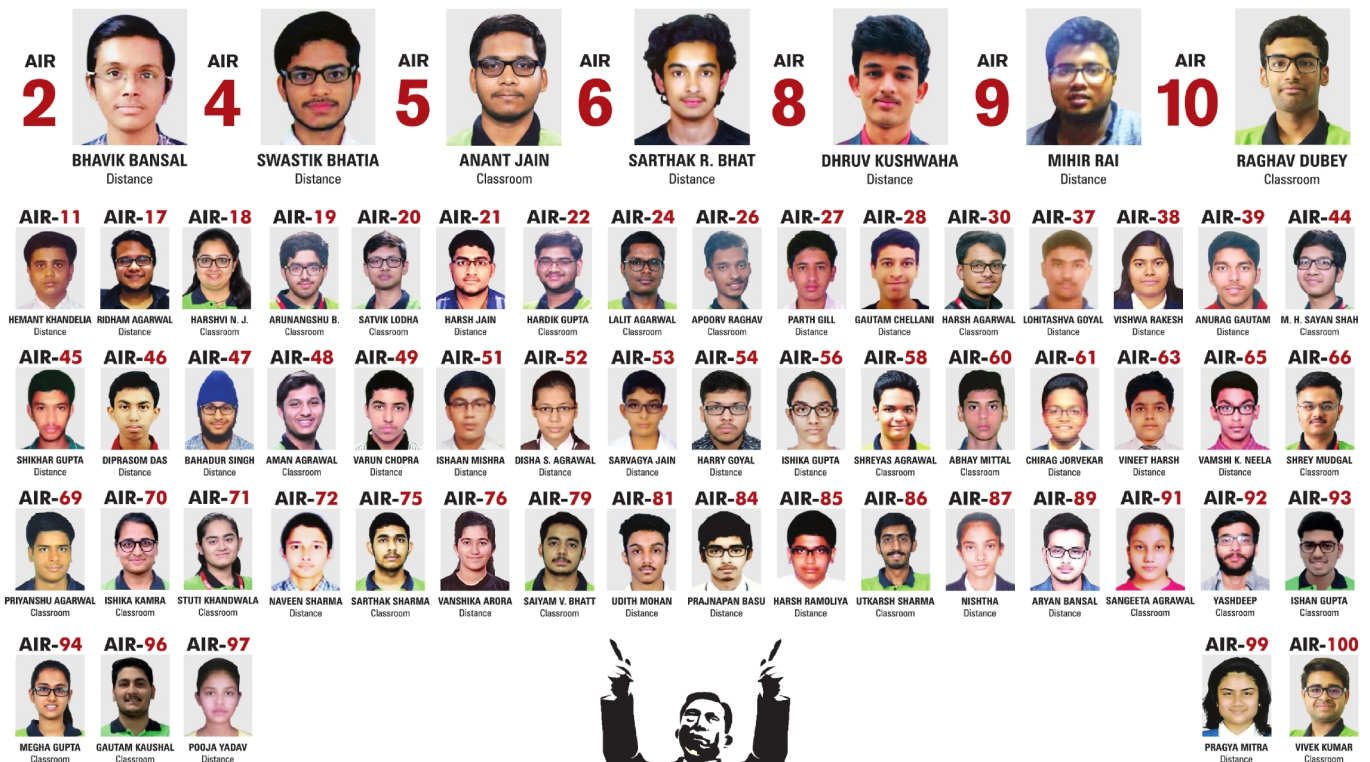
- 167.** Which of the following are called bacteria without cell wall
 (1) Virus
 (2) Viroid
 (3) Mycoplasma
 (4) Cyanobacteria
- 168.** *Penicillin* do not affect the mycoplasma because
 (1) There is no cell wall
 (2) There is no nucleus
 (3) There is no mitochondria
 (4) There is no golgi body
- 169.** The 'Witches broom' of legumes is caused by a
 (1) Virus
 (2) Mycoplasma
 (3) Bacterium
 (4) Fungus
- 170.** Nitrogenase enzyme is found in *Nostoc* in the cell of
 (1) Vegetative
 (2) Heterocyst
 (3) Both (1) and (2)
 (4) None of these
- 171.** The characteristic of blue green algae is
 (1) DNA without histone
 (2) Nuclear membrane absent
 (3) 70 S ribosomes
 (4) All of the above
- 172.** The blue-green algae are so called as they have in addition to green pigment chlorophyll, a blue pigment known as
 (1) Phycocyanin
 (2) Chromoplasm
 (3) Cyanophycin
 (4) Phycoerythrin
- 173.** Cyanobacteria are
 (1) Mosses which attack bacteria
 (2) Bacteria which attack cyanophyceae
 (3) Autotrophic organism with phycocyanin
 (4) None of these
- 174.** Some protists possess structures for regulation of their water content. They are
 (1) Nuclei
 (2) Contractile vacuoles
 (3) Chromatophores
 (4) Membranes
- 175.** Characteristic spores of diatoms are
 (1) Ascospores
 (2) Basidiospores
 (3) Auxospores
 (4) Zoospores
- 176.** *Ceratium* is
 (1) Dinoflagellate
 (2) Diatom
 (3) Slime mould
 (4) Sporozoan
- 177.** Murein is not found in the cell wall of
 (1) *Nostoc*
 (2) Eubacteria
 (3) Cyanobacteria
 (4) Diatoms
- 178.** Which one show bioluminescence
 (1) *Noctiluca*
 (2) *Polystomella*
 (3) *Entamoeba*
 (4) *Suctorina*
- 179.** Protists are
 1. Unicellular and prokaryote
 2. Unicellular and eukaryote
 3. Multicellular and eukaryote
 4. Autotroph and heterotroph
 (1) 1 + 2 + 3 (2) 2 + 3 + 4
 (3) 3 + 4 (4) 2 + 4
- 180.** Which one of the following fixes CO₂ in to carbohydrates :-
 (1) *Rhizobium*
 (2) *E.coli*
 (3) *Bacillus*
 (4) *Rhodospirillum*

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